Optimal streamlined bodies and the practical limits of tail-mounted boundary-layer-ingesting propulsion: a reduced-order and RANS study

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Abstract

We revisit two classical questions about bodies moving through a fluid at constant speed. First, which axisymmetric shape minimises the drag at fixed volume or at fixed frontal area? Second, how much propulsive power can be saved by placing the propulsor at the tail so that it ingests the body's boundary layer (boundary-layer ingestion, BLI, as in the Goldschmied body), and how large may the duct losses be before the saving disappears? A two-dimensional lattice-Boltzmann model, an axisymmetric panel/integral-boundary-layer model with shape optimisation, and axisymmetric RANS simulations (k–ω SST, OpenFOAM) are combined. The optimum is a conventional elongated teardrop, with fineness ratio $L/D \approx 3$ at fixed frontal area and $L/D \approx 6$ at fixed volume; with a turbulent boundary layer, axisymmetric golden-ratio bodies have 2.4–3.6 times the drag of the optimum (a sphere 3.8–7.5 times), in agreement with Hoerner's classical data and with earlier optimisation studies. For a fully turbulent boundary layer the reduced-order model gives a break-even duct loss K^* that decreases from 8.5% to 4.5% of the free-stream dynamic pressure as Re_V goes from 3×10^5 to 10^8. RANS simulations at $Re_V = 10^6$ (and 10^7 for selected cases) confirm the BLI gain: an unducted actuator disc ingesting the boundary layer needs 8% less power than the bare body pushed by an ideal propeller of infinite disc area, and 15% less than with an isolated propeller of the same disc area. A flush wall slot loses essentially all the ingested total pressure ($K \approx 0.8$–1.0). A well-designed ducted intake with a rounded lip keeps the intake loss at $K \approx 2\%$, yet its self-propelled power ranges from 2% below to 9% above that of the conventional reference, depending on the size of the reference propeller; the best of four cowls tested (shorter and thinner) lowers this to between -6% and $+4\%$: the cowl skin friction, the altered afterbody pressure and the acceleration of the ingested flow cancel most or all of the BLI benefit. The ideal BLI gain and realistic installation penalties are of the same order, which explains the mixed experimental record of Goldschmied-type bodies. All codes and case generators are provided.

1 Introduction

Two questions motivate this note. (i) Is there a "special" body shape — for instance one based on the golden ratio or on a logarithmic spiral — that moves through a fluid with less resistance than the classical streamlined body? (ii) Can the propulsive power be reduced by re-energising the slow fluid of the body's own boundary layer with a propulsor placed at the tail, instead of accelerating undisturbed fluid with a conventional propeller?

The first question has a well-known answer in the engineering literature [1], and minimum-drag axisymmetric bodies have been optimised with panel/boundary-layer methods before [2, 3, 4]; we re-derive it with reproducible tools and quantify how far the "golden" shapes are from the optimum. The second question is the basis of the Goldschmied body [14] and of modern BLI aircraft concepts [16, 17, 18]. Its theoretical benefit is clear, but experimental results for suction bodies have been contradictory [15]. Our aim is to separate the ideal BLI gain from the

installation losses and to express the result as a simple design criterion: the maximum tolerable duct loss K^*.

2 Methods

2.1 Two-dimensional lattice-Boltzmann screening

A D2Q9 lattice-Boltzmann solver [13] (`lbm.py`) was used at $Re = 200$ (based on body height H, 30 lattice units) to compare a circle, golden-ratio shapes (ellipse, rectangle, teardrop, golden spiral) and teardrops of $L/H = 3$ and 4.5. This 2D step is a qualitative screening only.

2.2 Axisymmetric reduced-order model

The method follows the classical approach of [5, 3, 4] (`axi.py`): the inviscid flow is computed with an axisymmetric panel method (vortex rings on the surface); the laminar boundary layer with Thwaites' method [6] in its axisymmetric form; transition with the e^N envelope method of Drela and Giles [7] or with Michel's criterion [8]; the turbulent boundary layer with Head's entrainment method [9] and the Ludwieg–Tillmann skin-friction law [10]; separation is detected from Thwaites' parameter ($\lambda < -0.09$, laminar) or the shape factor ($H > 2.4$, turbulent), and a separated afterbody receives a base-drag estimate. The total drag follows from Young's formula [5] applied to the momentum thickness near the tail. Bodies are described by five parameters (fineness ratio and four class/shape-function coefficients [11]). The objective is the volumetric drag coefficient $C_{D,V} = D/(\frac{1}{2}\rho U^2 V^{2/3})$ (fixed volume) or the drag based on frontal area. Optimisation uses differential evolution [12] (`axi_opt.py`). To test how much a shape designed for natural transition depends on the uncertain transition location, six scenarios are defined (`robusta.py`): e^N transition with free-stream turbulence $Tu = 0.05$, 0.1, 0.3 and 1%, Michel's criterion, and a "rough" case with transition forced at $Re_x = 2 \times 10^6$; transition is in any case capped at $Re_x = 5 \times 10^6$ (10^7 for $Tu = 0.05\%$). At fixed volume and power the speed scales as $C_{D,V}^{-1/3}$. Three shapes are compared at each Re_V: the turbulent optimum (reference), the *nominal* optimum for $Tu = 0.1\%$, and a *guaranteed* shape that maximises the smallest speed gain over the six scenarios.

2.3 BLI model and break-even loss

The tail propulsor is modelled as an ingestion slot at x_s with volume flow Q (`bli.py`). The pump raises the total pressure of the ingested flow so that the jet leaves at free-stream total pressure plus the momentum needed for self-propulsion. Duct losses are expressed as a total-pressure loss $K\,\frac{1}{2}\rho U^2$. The same blade efficiency $\eta = 0.85$ is used for the pump and for the reference propeller, so the comparison isolates the BLI physics and the duct loss. For each Re_V and $K \in \{0, 0.05, 0.10, 0.20\}$ the body shape, x_s and Q are optimised jointly (`bli_opt.py`, `soglia.py`); K^* is the loss at which BLI and conventional propulsion require the same power.

2.4 RANS simulations

Axisymmetric RANS simulations were run with OpenFOAM 14 (`foamRun`, incompressible, SIMPLEC, linear-upwind convection, k–ω SST [19]) on a 5° wedge (case generators `cfd_caso.py`, `cfd_blockmesh.py`, `cfd_gold.py`, `cfd_carena.py`). The body is the turbulent optimum at $Re_V = 10^6$ ($Re_L = 4.95 \times 10^6$, $L/D = 6.47$). Lengths are scaled by L, velocities by U, forces are reported per wedge in units of $\rho U^2 L^2$. The wall is resolved without wall functions: on the body the area-averaged y^+ of the first cell is 0.32–0.43 and the maximum 0.54–0.89 in all cases at $Re_V = 10^6$ (at 10^7 the first cell was rescaled to keep the same y^+, not measured); on the cowl of the ducted case the average is 1.45 and the maximum 2.5. The grid refinement of Sec. 3.9

increased the cell count in all directions but kept the first-cell height, so y^+ is unchanged on the fine grids; a 5% error on the cowl skin friction would change the ducted power by about 0.8 points.

Inflow turbulence and tripping. With low free-stream turbulence ($k = 1.5 \times 10^{-6}$, $\omega = 10$) the SST model stayed laminar over the whole body and produced a spurious 45% power saving. Unless stated otherwise, all cases reported here use $\omega_\infty = 1.5$ ($\nu_t/\nu = 5$) and a trip region ($k = 0.005$, $\omega = 100$ for $0.03 < x < 0.08$). The eddy-viscosity distribution along the body was checked against the bare-body reference in every case (`nut_x.py`).

Propulsor. The fan is an actuator disc (momentum source) of axial thickness $0.01L$. Self-propulsion is reached iteratively: the disc thrust T is set equal to the net drag of body and cowl, the flow is re-converged, and the procedure is repeated three times (`Allrun_std`). The shaft power of an ideal disc is $P = T\,\langle u\rangle_{\mathrm{disc}}$; it includes the kinetic energy left in the jet. It is compared with the bare body pushed by an isolated (non-interacting) propeller in free stream, modelled as an actuator disc of area A_{ref} with $T = D_{\mathrm{bare}}$:

$$P_{\mathrm{conv}} = D_{\mathrm{bare}}\,U\,\frac{1+\sqrt{1+C_T}}{2}, \qquad C_T = \frac{D_{\mathrm{bare}}}{\frac{1}{2}\rho U^2 A_{\mathrm{ref}}}. \tag{1}$$

with drag and area taken over the full annulus. The limit $A_{\mathrm{ref}} \to \infty$ gives $P_{\mathrm{conv}} = D_{\mathrm{bare}}U$ (ideal propeller, no jet loss), the most favourable reference for conventional propulsion; unless stated otherwise, power changes are quoted against this limit, and Table 6 gives the sensitivity to A_{ref} (`confronto_elica.py`).

Intake loss. K is the difference between the mass-averaged total pressure of the captured stream tube at $x = 0.835$ and at the fan face, divided by $\frac{1}{2}\rho U^2$ (`misura_K.py`, `misura_carena.py`).

Configurations: (a) bare body; (b) free actuator disc at $0.895 \le x \le 0.905$, radius 0.03; (c) flush wall suction slot (Goldschmied type), slot widths 0.006–0.024; (d) ducted intake with an annular NACA-type cowl (leading edge at $x = 0.855$, chord 0.095, thickness 5%, mean radius $0.0293 \to 0.0249$; intake, fan and exit areas 2.54, 1.91, $1.89 \times 10^{-3}\,L^2$) with the disc inside the duct. The equal-area references below use the effective disc area $Q/\langle u\rangle$: 2.79×10^{-3} for the free disc (geometric 2.83×10^{-3}) and 1.94×10^{-3} for the ducted fan (geometric 1.91×10^{-3}). Three further cowls change one parameter each (`genera_v3.py`): (e) chord 0.070 and thickness 4%; (f) mean radius $0.0293 \to 0.0275$ (less convergent duct); (g) $0.0293 \to 0.0290$ (nearly constant section). A second Reynolds number, $Re_L = 4.95 \times 10^7$ ($Re_V \approx 10^7$), was computed for the bare body, the free disc and cowl (d) on the same body, with the first-cell height rescaled for $y^+ < 1$ and $\omega_\infty = 15$ to keep $\nu_t/\nu = 5$ at the inflow.

3 Results

3.1 Optimal body shape

In 2D at $Re = 200$ (Fig. 1) the area-based drag coefficient of the golden-ratio shapes is 1.08–1.39 (circle: 1.79), against 0.73 for the $L/H = 3$ teardrop and 0.566 for the $L/H = 4.5$ teardrop, i.e. 1.9–2.5 times the best 2D shape. (A 2D shape optimisation was also run, but at a coarser lattice than the comparison shapes; its result is not comparable and is not reported.)

In axisymmetric flow (Table 1, Fig. 2) the optimum at fixed volume and turbulent boundary layer has $L/D = 6.6$ ($Re_V = 10^7$). At fixed frontal area the optimum is $L/D = 3.2$. With a turbulent boundary layer, golden-ratio bodies ($L/D = 1.618$) separate on the afterbody and have 2.4–3.6 times the drag of the optimum; a sphere has 3.8–7.5 times. These values are consistent with Hoerner's compilation [1] and with earlier optimisations [2, 4].

The natural-transition rows of Table 1 were computed with Michel's criterion and the cap $Re_x \le 5 \times 10^6$; they give $L/D = 6.0$ at $Re_V = 10^7$ and $L/D = 9.3$ at $Re_V = 10^6$, the latter with laminar flow up to $x/L = 0.87$. They are optimistic and should be read as an upper bound:

re-evaluated with the e^N criterion ($Tu = 0.1\%$), the same $Re_V = 10^6$ body has $C_{D,V} = 0.0180$ instead of 0.0072, while the $Re_V = 10^7$ body is unchanged (0.0165). The effect of the transition criterion on the shape is quantified in the next section, where the nominal e^N optimum at $Re_V = 10^6$ has $L/D = 5.6$.

Table 1: Axisymmetric shape optimisation (reduced-order model). C: drag coefficient on the optimisation reference (volume$^{2/3}$ or frontal area). "Natural": transition by Michel's criterion, $Re_x \leq 5 \times 10^6$ (optimistic, see text).

Case	L/D_{opt}	C_{opt}	gold. ellipsoid	gold. teardrop	sphere	$L/D = 4.5$
Volume, turbulent, $Re_V = 10^7$	6.58	0.0172	0.0617	0.0530	0.1288	0.0175
Volume, natural, $Re_V = 10^7$	6.00	0.0165	0.0609	0.0516	0.1279	0.0168
Volume, natural, $Re_V = 10^6$	9.29	0.0072	0.0589	0.0540	0.1268	0.0133
Frontal area, turb., $Re_D = 10^6$	3.17	0.0370	0.0967	0.0898	0.1400	0.0506

3.2 Robustness to the transition location

Table 2 and Fig. 3 give the speed change at equal power relative to the turbulent optimum. At $Re_V = 10^6$ the nominal shape ($L/D = 5.6$) is 16–18% faster in calm conditions but 6–7% slower when the free-stream turbulence is 0.3% or more or the surface is rough: its gain depends on a long laminar run that disturbances destroy. The guaranteed shape ($L/D = 4.0$) is faster in every scenario, but only by 2.7–3.6%. At $Re_V = 3 \times 10^6$ the guaranteed gain is 0.4%, and from $Re_V = 10^7$ on all shapes coincide with the turbulent optimum within 0.6%, because the laminar run is a small fraction of the length. These numbers are optimistic in three respects: in some scenarios transition is triggered by laminar separation, with no penalty for the separation bubble; the cap on the laminar run is an assumption that decides the result at high Re_V; and the nominal optimum at $Re_V = 3 \times 10^6$ ($L/D = 2.9$) is probably a local minimum. The robust conclusion is the qualitative one: a laminar-flow shape can gain much or lose much, and the gain that can be guaranteed is small.

Table 2: Speed change at equal power relative to the turbulent optimum, range over the six transition scenarios (reduced-order model, `robusta.py`).

Re_V	nominal L/D	nominal shape	guaranteed L/D	guaranteed shape
10^6	5.6	-6.9 to $+17.7\%$	4.0	$+2.7$ to $+3.6\%$
3×10^6	2.9	-4.3 to $+12.0\%$	5.3	$+0.4$ to $+4.2\%$
10^7	6.0	-0.5 to $+0.6\%$	6.5	0.0 to $+0.2\%$
3×10^7	6.5	0.0%	6.5	0.0%

3.3 Break-even duct loss K^*

For a fully turbulent boundary layer the ideal BLI saving ($K = 0$) is 17–20% of the power of the optimal conventional body. The saving drops steeply with K (Table 3, Fig. 4); the break-even loss K^* decreases from 8.5% at $Re_V = 3 \times 10^5$ to 4.5% at 10^8, because the boundary layer becomes relatively thinner and carries less momentum deficit. At $Re_V = 10^7$, three repeated optimisations with different seeds and more iterations gave $K^* = 5.83$, 5.85 and 5.88% (table value 5.77%), i.e. a spread of about 0.1 percentage points; the repeatability was not tested at the other Reynolds numbers. The reference here is the ideal propeller ($P = D\,U/\eta$); the joint optimisation leaves the ingested flow rate Q free, so the BLI jet loss is also small. A finite-area reference (Eq. 1) would raise K^*; the values in Table 3 are therefore conservative. For natural

transition the three transition criteria tested (Michel, e^N, Langtry–Menter [20]) give widely different laminar extents, and K^* is not determined reliably; we do not report it as a result (the shaded band in Fig. 4 only shows this spread).

Table 3: Power change of the BLI body relative to the optimal conventional body (turbulent boundary layer), and break-even loss K^*.

Re_V	$K = 0$	$K = 0.05$	$K = 0.10$	$K = 0.20$	K^*
3×10^5	−20.1%	−6.5%	+2.9%	+15.9%	8.5%
10^6	−19.8%	−4.8%	+4.7%	+18.1%	7.5%
3×10^6	−19.0%	−3.1%	+6.5%	+19.9%	6.6%
10^7	−18.1%	−1.6%	+8.6%	+22.1%	5.8%
3×10^7	−17.3%	0.0%	+10.4%	+23.6%	5.0%
10^8	−16.9%	+1.8%	+11.9%	+25.0%	4.5%

Validation against Goldschmied. For Goldschmied's $L/D = 3$ body at $Re_L = 10^7$ ($Re_V = 3.7 \times 10^6$) the model gives $C_{D,V} = 0.0172$ with ideal suction against 0.0213 for the best conventional $L/D = 3$ body (−19%). Goldschmied reported 0.0162 versus 0.0235 for the Akron-type reference (−31%; −26% in total engine power) [14]. The model therefore predicts a smaller, not a larger, benefit than the original claim.

3.4 RANS: bare body and free actuator disc

The bare-body drag is 6.957×10^{-6} per wedge, 4.6% above the reduced-order model, with matching pressure distribution (Fig. 5). With the free actuator disc in self-propulsion, the ideal shaft power is 8.2% lower than P_{conv} for an ideal propeller, and 15.3% lower than for an isolated propeller of the same disc area ($A = 2.79 \times 10^{-3} L^2$, Table 6). The disc ingests the inner part of the boundary layer (mean axial velocity at the disc $0.89\,U$), the required thrust is $T/D_{\mathrm{bare}} = 1.03$ (the disc raises the body drag by 3%), and the measured intake loss is $K \approx 0 \pm 1\%$. This is the ideal BLI benefit at this Reynolds number within RANS; 8.2% is its lower bound.

3.5 Comparison with the Delft experiment

Della Corte et al. [18] measured with PIV the power balance of an axisymmetric fuselage ($L/D \approx 11$, boundary layer tripped at 5% of the length) with and without an aft-mounted shrouded fan; the PIV used for the power balance was taken at 20 m/s, $Re_L \approx 2.4 \times 10^6$ (campaign 1.7–4.7×10^6). For the bare fuselage they found that the mechanical energy deposited in the flow at the trailing edge, $C_{\dot{E}}$ (their Eq. 10: axial, radial and tangential kinetic energy plus pressure work, $\dot{E}_a + \dot{E}_v + \dot{E}_p$ in Drela's notation [17]), is about 13% of the momentum-deficit power $F_x U$ at the same station; this is the ideal BLI benefit, the part of the losses a propulsor could still recover. The same quantity from our bare-body RANS solution at $Re_L = 4.95 \times 10^6$, about twice the experimental value (`delft_confronto.py`; momentum balance closed to within 3%), is 9.9% at the trailing edge: the kinetic terms give 11.7% of DU and the pressure work −2.1% (Table 4). The experimental value is higher, as expected for a lower Reynolds number and a relatively thicker boundary layer, and the two agree in order of magnitude on different bodies ($L/D = 6.5$ against 11). Two observations agree qualitatively: the operating fan increases the momentum deficit upstream of the shroud by about 10% in the experiment, and our body drag rises by about 3% with the free disc and 4% with the cowl (Table 5); and in the experiment about 30% of δ_{99} is not ingested, while in our free-disc case the outer part of the boundary layer also passes outside the disc. Our free disc recovers 8.2% of the 9.9% available, i.e. most of it; the saving against an equal-area propeller (15%) can exceed the wake energy because it also contains the jet loss of the reference propeller.

Table 4: Mechanical energy deposited in the flow behind the bare body (kinetic terms and pressure work, as in Della Corte et al.'s Eq. 10), at the trailing edge unless stated: RANS (this work) and experiment [18].

	L/D	Re_L	$(\dot{E}_a + \dot{E}_v)/(DU)$	$\dot{E}/(F_x U)$
Della Corte et al. (PIV)	≈ 11	$\approx 2.4 \times 10^6$	–	$\approx 13\%$
This work (RANS, $x = 1.0$)	6.5	4.95×10^6	11.7%	9.9%
This work (RANS, $x = 1.05$ / 1.2)	6.5	4.95×10^6	10.8% / 9.1%	9.7% / 8.8%

3.6 RANS: flush wall slot

A flush suction slot gives $K \approx 0.84$–1.02 for slot widths 0.006–0.012 (and 1.28 for 0.024). The no-slip wall condition across the slot removes the tangential velocity of the ingested fluid: the slot behaves as a plenum intake that discards the whole axial kinetic energy of the captured stream. Such an intake is far above K^* and cannot yield a net saving. (The slot cases were run with a prescribed jet velocity, not in self-propulsion, on a different body (the natural-transition BLI optimum); their pump power is therefore not reported as a power saving or penalty.) This is qualitatively consistent with the Cal Poly wind-tunnel results [15].

3.7 RANS: ducted intake with a rounded lip

A first cowl design (intake area 3.4×10^{-3}, lip thickness 8%) produced separation and reversed flow ahead of the intake, with oscillating forces: the intake was too large for the available boundary-layer flow. The final design (Sec. 2.4, cowl (d)) is attached (Fig. 6) and has $K = 2.1\%$, well below K^*. Nevertheless the self-propelled power is 9.2% *higher* than P_{conv} for an ideal propeller. Against an isolated propeller of the same area as the fan ($A = 1.94 \times 10^{-3} L^2$) it is 2.1% lower, and against a propeller as large as the free disc it is 0.9% higher (Table 6). The ducted intake therefore gives no clear net gain. Table 5 decomposes the difference relative to the bare-body drag.

Table 5: Force decomposition of the ducted configuration in self-propulsion, relative to the bare-body drag $D_{\mathrm{bare}} = 6.957 \times 10^{-6}$.

Cowl skin friction	+15.4%
Lip suction (leading-edge thrust)	−11.4%
Body drag increase (afterbody suction)	+4.0%
Required thrust T/D_{bare}	1.080
Mean axial velocity at the fan	$1.01\,U$ (free disc: $0.89\,U$)
Power P/P_{conv} (ideal propeller)	1.092

A fairer reference for the ducted intake would be an isolated ducted fan (nacelle) of the same size, which also carries cowl friction; it has not been computed.

The duct accelerates the ingested flow to $1.01\,U$, so the fan no longer works on slow fluid and the BLI benefit disappears. A simple estimate with the same cowl and a fan velocity of $0.89\,U$ gives $P/P_{\mathrm{conv}} \approx 0.96$ (−4% against the ideal propeller).

Other cowls. Table 7 compares the four cowls (`misura_v3.py`; ν_t along the body turbulent in all cases). The shorter and thinner cowl (e) is the best: the fan sees $0.97\,U$ instead of $1.01\,U$ and the power drops by 5 points, to +4.3% against the ideal propeller and −6.2% against an equal-area one. Opening the exit, cowls (f) and (g), has the opposite effect to the one intended: the convergent exit acts as a throttle, and widening it draws more flow through the duct and accelerates the fan inflow to 1.10–$1.16\,U$, so the power rises to +12–15%. To make the fan see slower fluid the exit must be narrower, or the fan larger for the same exit (not computed). The

Table 6: Self-propelled power of the BLI configurations relative to the bare body pushed by an isolated propeller of disc area A_{ref} (Eq. 1; base grid). Fine grid, equal-area references: free disc -15.2%, ducted -1.4% to -1.7%.

Reference propeller	A_{ref}/L^2	Free disc	Ducted intake
same area as the ducted fan	1.94×10^{-3}	-17.8%	-2.1%
same area as the free disc	2.79×10^{-3}	-15.3%	$+0.9\%$
twice the free-disc area	5.58×10^{-3}	-12.0%	$+4.7\%$
half the body frontal area	9.37×10^{-3}	-10.6%	$+6.4\%$
body frontal area	1.87×10^{-2}	-9.4%	$+7.8\%$
infinite (ideal propeller)	∞	-8.2%	$+9.2\%$

intake loss stays at 1.8–3.1% for all cowls, well below K^*: the penalty comes from cowl friction and fan inflow speed, not from the intake.

Table 7: Ducted configurations in self-propulsion at $Re_V = 10^6$ (base grid). Power relative to the bare body with an ideal propeller and with an isolated propeller of the same effective disc area A (Eq. 1).

Cowl	K	fan inflow	T/D_{bare}	$A \times 10^3$	vs ideal	vs equal area
(d) reference	2.1%	$1.01\,U$	1.080	1.94	$+9.2\%$	-2.1%
(e) chord 0.070, 4%	1.8%	$0.97\,U$	1.080	2.00	$+4.3\%$	-6.2%
(f) exit radius 0.0275	2.7%	$1.10\,U$	1.016	2.12	$+12.1\%$	$+1.2\%$
(g) exit radius 0.0290	3.1%	$1.16\,U$	0.991	2.22	$+14.9\%$	$+4.3\%$

3.8 Effect of Reynolds number

At $Re_L = 4.95 \times 10^7$ the bare-body drag per wedge falls to 4.911×10^{-6} (-29%). The free disc saves 6.7% of the power against an ideal propeller and 11.9% against an equal-area propeller, against 8.2% and 15.3% at $Re_L = 4.95 \times 10^6$: the boundary layer is relatively thinner and carries less momentum deficit, the same trend as K^* in Table 3. The ducted case (d) did not converge at this Reynolds number: with a damped thrust update the thrust–drag mismatch stayed at 2–3% and the drag kept falling by about 1.4% every 800 iterations, so the flow itself had not settled; it is not reported. Its eddy viscosity along the body matches the bare body at the same Reynolds number but is three times lower at $x = 0.75$ than in a first, undamped attempt, a reminder of the bistability of the SST solution (Sec. 3.9). The bare-body and free-disc values at this Reynolds number are converged (drag change below 0.01% over the last 400 iterations) and turbulent along the body.

3.9 Grid independence

On the refined grid the drag and the self-propelled powers change by less than one percentage point, and the intake losses by at most 1.1 points (Table 8). A third, coarser grid (factor $1/\sqrt{2}$) gives a monotonic sequence for the bare-body drag, and a grid error of at most 0.5% (grid convergence index with safety factor 1.25 [21]). Over the last 400 iterations the drag changed by less than 0.01% on the coarse and base grids but by -0.15% on the fine grid (`deriva_v3.py`), with about 0.1% still to go; the observed order (between 1.2 and about 4.7 depending on the final fine-grid value) and the Richardson estimate (6.963–7.001×10^{-6}) are therefore indicative only, while the bound of 0.5% holds for all plausible values, and is smaller than the differences between configurations discussed in this note. The free-disc power changes non-monotonically within one point (-7.3, -8.2, -8.1%). The ducted case diverged on the coarse grid, whose cells

around the cowl are of poorer quality, and is not reported; for it only the base and fine grids are available, and the fine-grid ducted drag was still falling by 0.13% over its last 400 iterations, so its power could drop by a further 0.1–0.2 points. Two fine-grid runs (the ducted case with the standard inflow settings, and the bare body with $\omega_\infty = 0.3$ and a longer trip) converged instead to a spurious laminar state: the eddy viscosity decayed along the body and, in the ducted case, the laminar boundary layer separated at $0.54 < x < 0.70$. The SST solution is therefore bistable and depends on the grid and on the initial conditions; these runs were discarded after checking ν_t along the body. The fine ducted result reported here uses $\omega_\infty = 0.3$ and a trip extending to $x = 0.15$; its ν_t distribution along the body matches the base-grid case, and it is referred to the fine-grid bare body computed with the standard settings. Self-propulsion was not fully converged (thrust and drag differ by 0.3%); with $T = D$ the power change would be about +9.7%.

Table 8: Grid sensitivity (factor $\sqrt{2}$ per direction between grids; powers against the ideal propeller).

Quantity	coarse grid	base grid	fine grid	base→fine
Bare-body drag ($\times 10^{-6}$)	6.934	6.957	6.966–6.973	+0.2%
Free disc: $P/P_{\mathrm{conv}} - 1$	−7.3%	−8.2%	−8.1%	0.15 points
Free disc: K	−1.0%	0.5%	−0.6%	1.1 points
Ducted: $P/P_{\mathrm{conv}} - 1$	–	+9.2%	+9.7 to +10.0%	0.5–0.8 points
Ducted: K	–	2.1%	2.0%	0.1 points

4 Discussion

The ideal BLI benefit (8–15% in RANS for this body at $Re_V = 10^6$ depending on the reference propeller, 6.7–11.9% at 10^7; 17–20% in the reduced-order model, which optimises the shape jointly with the intake) is of the same order as the losses of a realistic installation. The difficulty is intrinsic: for the fan to see slow fluid the duct must diffuse, but a thick turbulent boundary layer that is further decelerated separates ahead of the intake; if instead the duct accelerates the flow, the intake is clean but the benefit is lost. The cowl itself adds wetted area whose friction is comparable to the benefit. This explains why careful experiments on suction bodies have struggled to reproduce the large savings originally reported.

Limitations. (1) Axisymmetric, steady RANS with a single turbulence model; no free-stream turbulence effects beyond the trip. (2) Ideal actuator disc: no blade losses, swirl or fan–distortion interaction. (3) Four cowl geometries were tested, each changing one parameter; no systematic cowl optimisation was made. (4) Laminar/transitional branch unresolved. (5) Two Reynolds numbers in RANS ($Re_V \approx 10^6$ and 10^7), on a single body; the ducted case did not converge at the higher one. (6) The SST solution can fall into a spurious laminar state (Sec. 3.9); every result was screened for it, but the screening is a posteriori. (7) The conventional reference is an isolated propeller that does not interact with the body; the power changes depend on its assumed size (Table 6). (8) The grid study uses three grids (two for the ducted case) with the same first-cell height; no wall-resolution study was made. (9) The only comparison with experiment (Sec. 3.5) concerns the bare-body wake energy, on a different body; the powered configurations are not validated against measurements.

5 Conclusions

1. The minimum-drag body is a conventional teardrop ($L/D \approx 3$ at fixed frontal area, ≈ 6 at fixed volume); golden-ratio shapes have 2.4–3.6 times the drag in axisymmetric turbulent flow (1.9–2.5 times in the 2D screening).

2. Shapes optimised for a long laminar run are fragile: with natural transition at $Re_V = 10^6$ they can be 18% faster or 7% slower than the turbulent optimum depending on the free-stream disturbances; a shape designed for the worst case guarantees only about 3%, and nothing above $Re_V \approx 10^7$.

3. For turbulent boundary layers, tail BLI pays only if the duct loses less than $K^* = 4.5$–8.5% of the dynamic pressure (conservative with respect to the reference propeller; reduced-order model, not validated in RANS).

4. A flush wall slot loses almost all the ingested total pressure and cannot pay.

5. An unducted tail propulsor ingesting the boundary layer saves 8–15% of the power at $Re_V = 10^6$ (6.7–11.9% at 10^7), depending on the size of the conventional propeller it is compared with.

6. A clean ducted intake ($K = 2\%$) is achievable, but cowl friction and flow acceleration cancel most or all of the benefit (power between -2% and $+9\%$ of the conventional reference); a shorter, thinner cowl brings it to between -6% and $+4\%$, and widening the duct exit makes it worse. Recovering a clear net gain requires a fan that sees slow fluid without a long, high-friction cowl; this is the open design problem.

Use of AI tools

The codes, simulations, analysis and text of this note were developed with extensive assistance from Claude (Anthropic), an AI assistant, under the direction of the author, who checked the results and takes full responsibility for the content.

Code and data availability

All scripts (Python 3, OpenFOAM 14), the exported RANS fields and the result files are available at `https://github.com/buongiornopietro/tail-bli-propulsion` (archived version v1.0: doi:10.5281/zenodo.22949389; all versions: doi:10.5281/zenodo.22949388). Table 9 maps each figure and table to the script that produces it; the repository README gives the commands.

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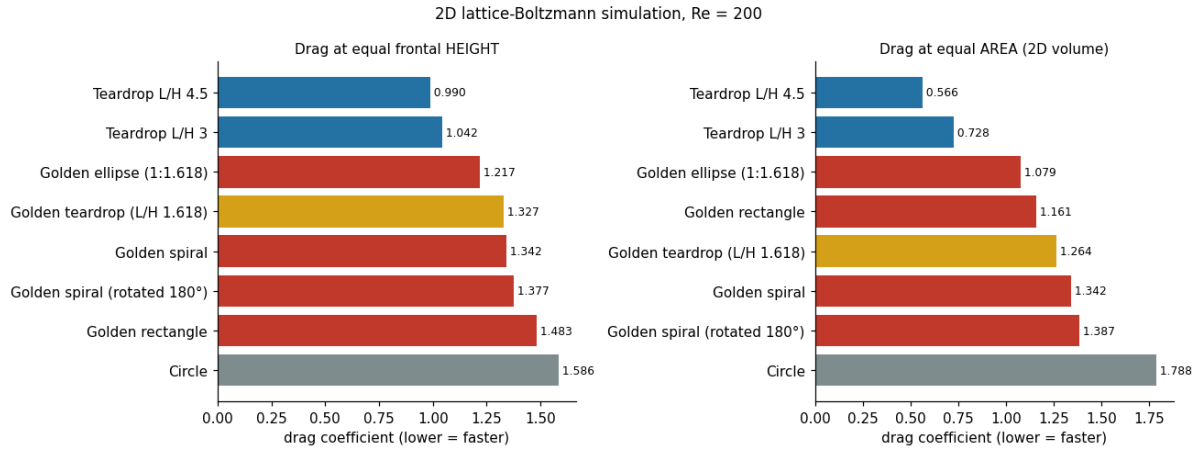
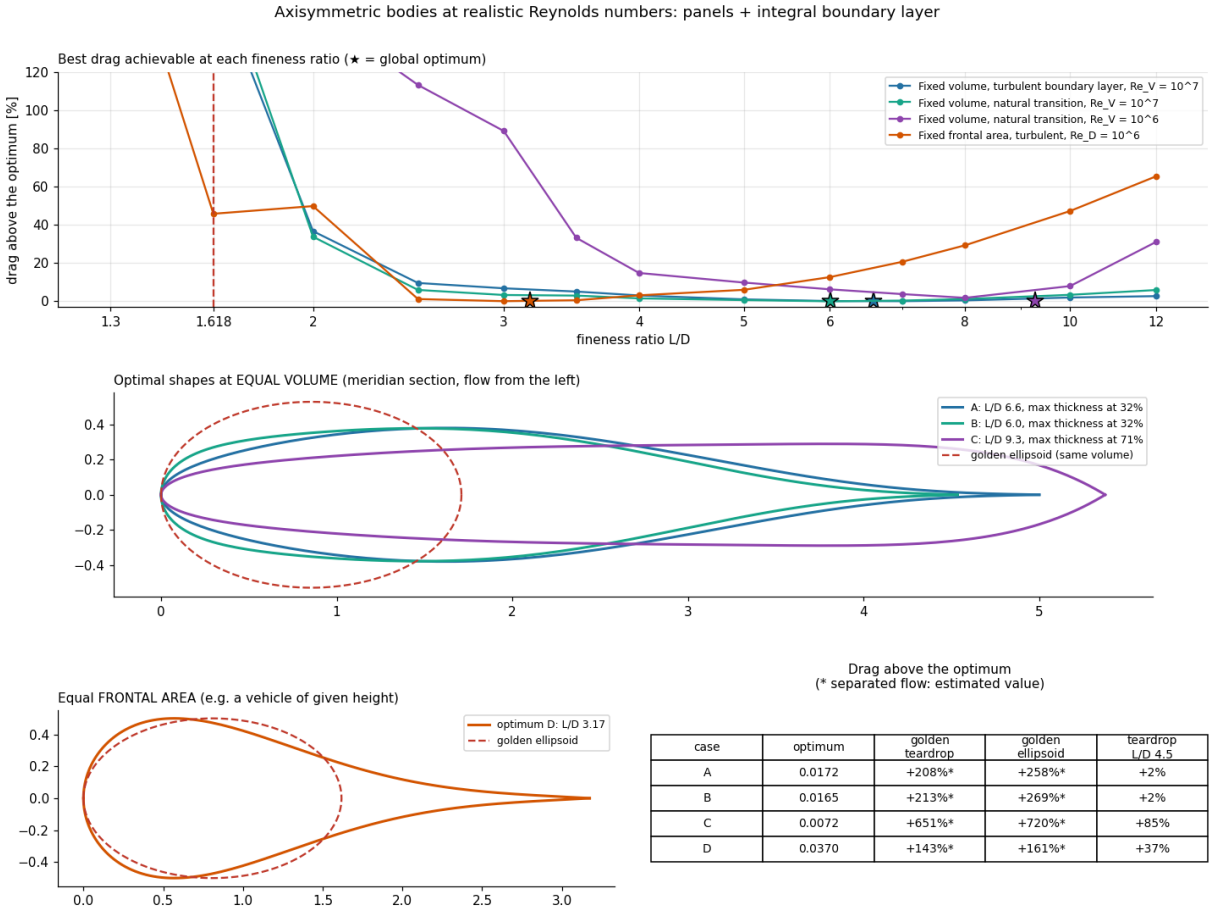


Figure 1: 2D lattice-Boltzmann drag at $Re = 200$ for golden-ratio shapes and teardrops.



Drag above the optimum
(* separated flow: estimated value)

case	optimum	golden teardrop	golden ellipsoid	teardrop L/D 4.5
A	0.0172	+208%*	+258%*	+2%
B	0.0165	+213%*	+269%*	+2%
C	0.0072	+651%*	+720%*	+85%
D	0.0370	+143%*	+161%*	+37%

Figure 2: Axisymmetric optimal bodies.

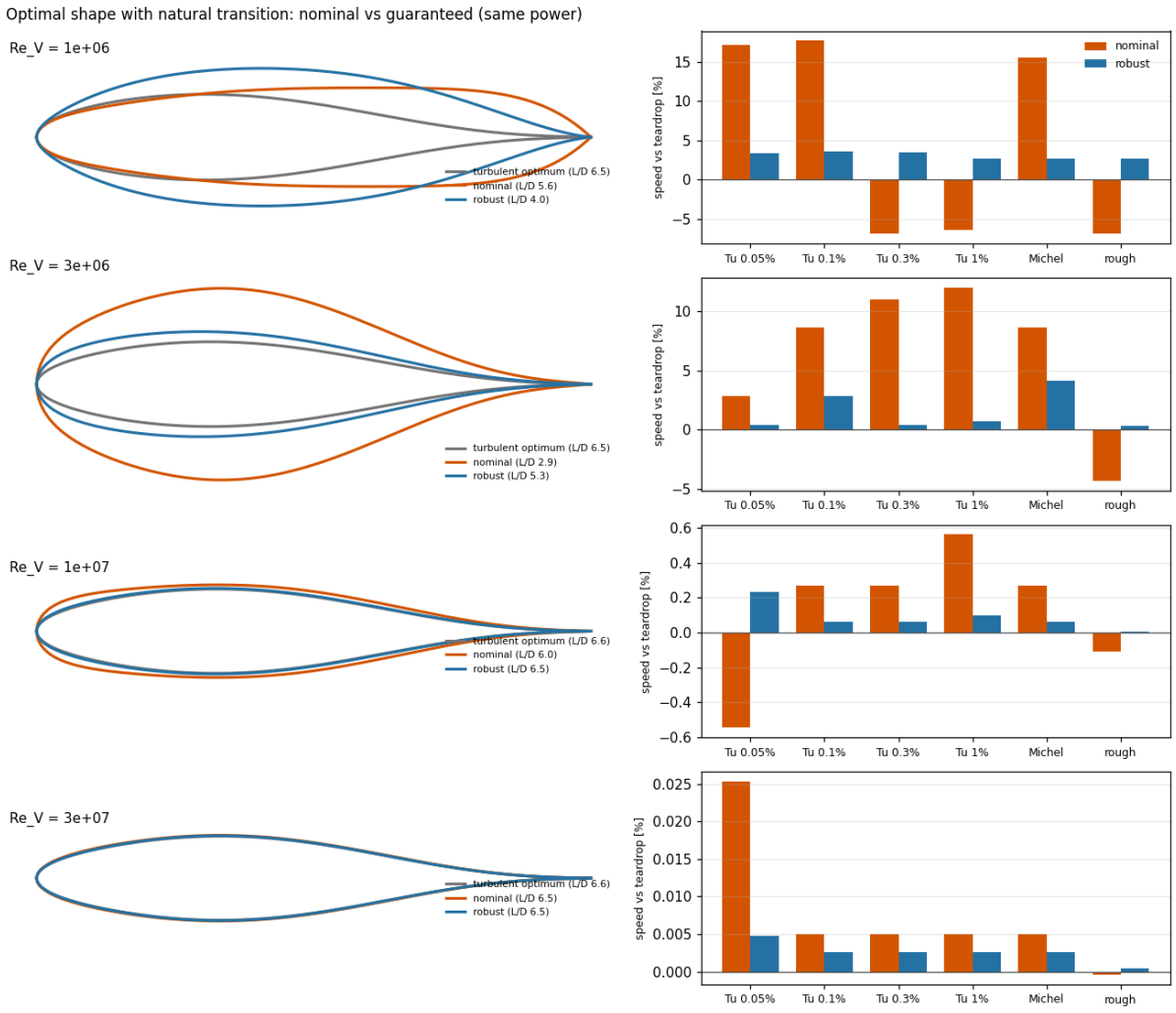


Figure 3: Nominal (optimised for Tu = 0.1%) and guaranteed (max–min) shapes with natural transition, and their speed change at equal power relative to the turbulent optimum in six transition scenarios.

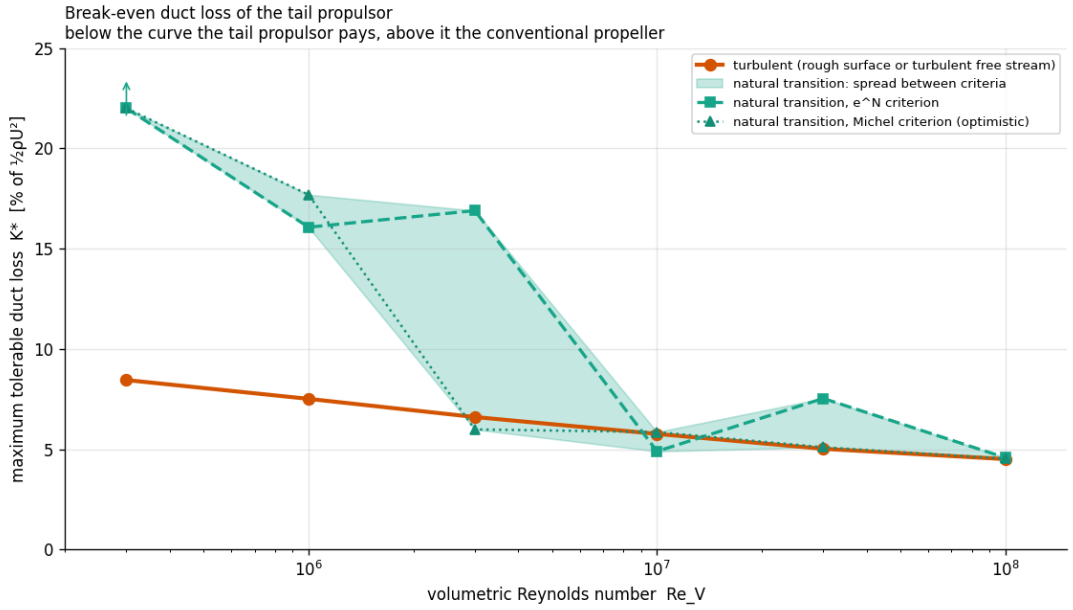


Figure 4: Break-even duct loss K^* versus Re_V. Solid line: fully turbulent boundary layer (result). Shaded band: spread between transition criteria for natural transition, shown for information only (not a result).

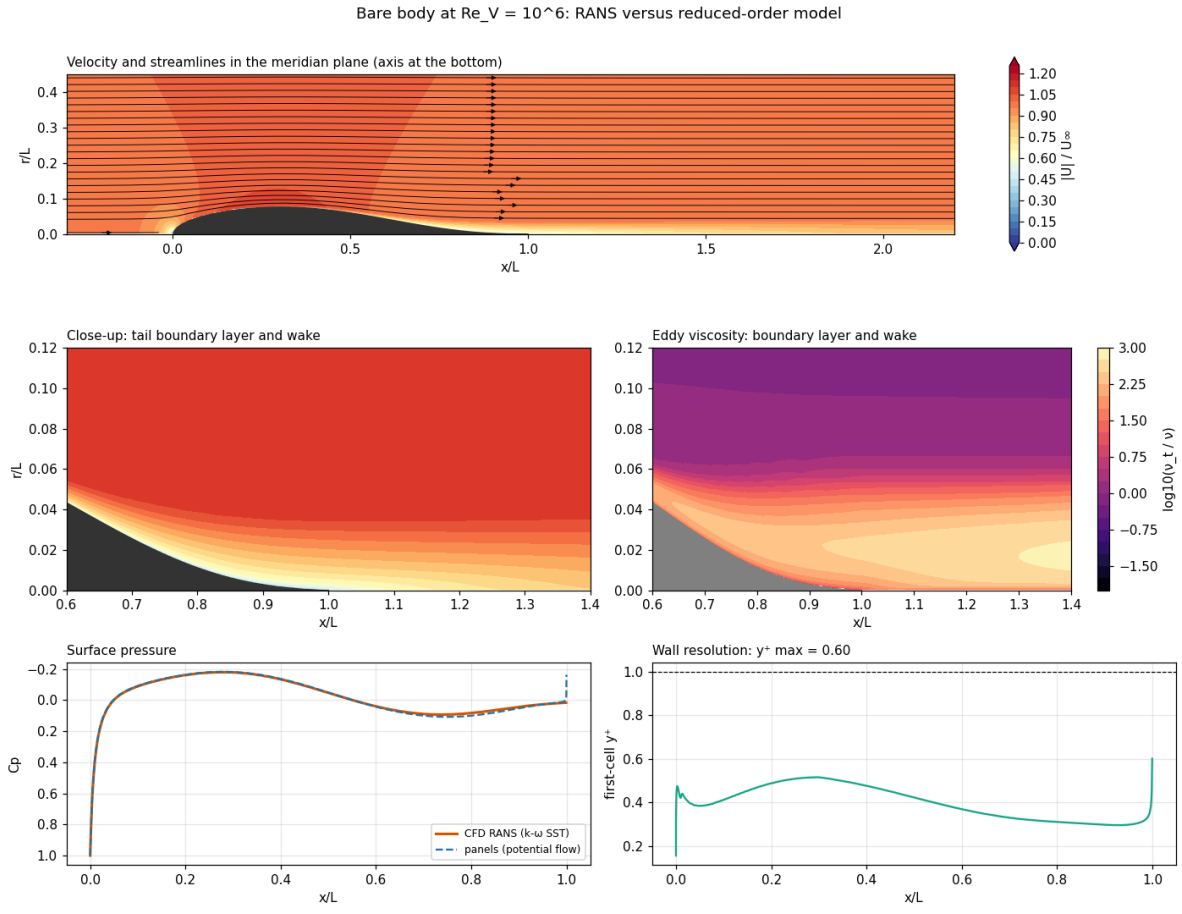


Figure 5: RANS versus reduced-order model, bare body at $Re_V = 10^6$.

Table 9: Scripts that produce each figure and table (all run from the repository root).

Item	Script (input → output)
Fig. 1	`lbm.py` → `plot.py`
Table 2, Fig. 3	`robusta.py` → `plot_robusta.py`
Table 1, Fig. 2	`axi.py`, `axi_opt.py` → `plot_axi.py`
Table 3, Fig. 4	`bli.py`, `bli_opt.py`, `soglia.py` → `plot_mappa_finale.py`
Goldschmied validation	`valida_goldschmied.py`
RANS cases	`cfd_caso.py`, `cfd_blockmesh.py`, `cfd_gold.py`, `cfd_carena.py`, `genera_fini.py`
Fig. 5	`plot_cfd.py`
Slot loss K	`misura_K.py`
Table 5, Fig. 6	`misura_carena.py`, `plot_carena.py`
Table 6	`confronto_elica.py`
Table 8	`misura_fine.py`
Table 4	`delft_confronto.py`
Table 7, Sec. 3.8, third grid	`genera_v3.py` (cases), `misura_v3.py`
Iterative drift (Sec. 3.9)	`deriva_v3.py`
ν_t check along the body	`nut_x.py`

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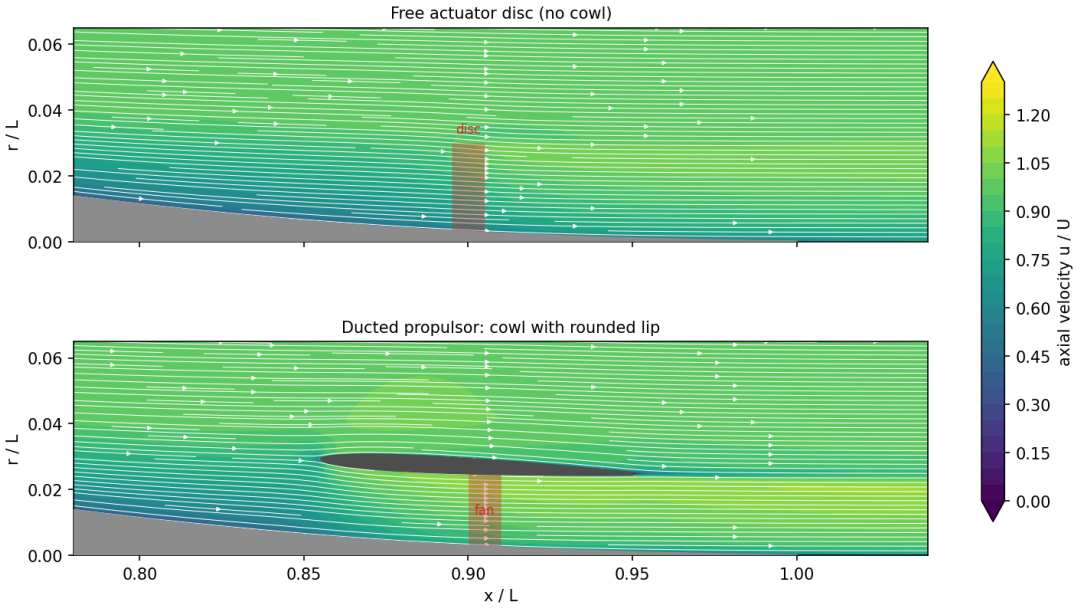


Figure 6: Flow near the tail with the ducted intake (cowl (d)).